THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 9: **Propositions and their Converses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The converse of the proposition "If a number is divisible by 6, then it is divisible by 3" is: 1

(a) If a number is divisible by 3, then it is divisible by 6.
(b) If a number is divisible by 6, then it is divisible by 3.
(c) If it is not true that a number is divisible by 6, then it is not true that it is divisible by 3.
(d) If it is not true that it is divisible by 3, then a number is divisible by 6.

2. The proposition "If a number is divisible by 6, then it is divisible by 3" is true. Its converse "If it is divisible by 3, then a number is divisible by 6" is: 1

(a) It cannot be decided **(b)** True
(c) True only for some values **(d)** False

3. Which of the following is a counterexample to the statement "If $a < b$, then $a^2 < b^2$"? 1

(a) $a = 1,\ b = 4$ **(b)** $a = 3,\ b = 6$ **(c)** $a = 2,\ b = 5$ **(d)** $a = -3,\ b = 2$

4. The number of factors of 36 is: 1

(a) 8 **(b)** 10 **(c)** 11 **(d)** 9

5. A triangle has sides 9 cm, 40 cm and 41 cm. Using the converse of the Baudhāyana–Pythagoras theorem, the triangle is: 1

(a) equilateral **(b)** not right-angled **(c)** right-angled **(d)** isosceles

6. For the expression $n^2 + n + 11$, the smallest natural number n for which the value is NOT prime is: 1

(a) 10 **(b)** 9 **(c)** 12 **(d)** 11

7. A proposition that is false can be shown to be false by giving: 1

(a) its converse **(b)** a counterexample
(c) a definition **(d)** many examples that satisfy it

8. Which of the following numbers has an odd number of factors? 1

(a) 25 **(b)** 24 **(c)** 20 **(d)** 27

9. The converse of the proposition "If a quadrilateral is a square, then it is a rectangle" is: 1

(a) If a quadrilateral is a rectangle, then it is a square.
(b) If it is not true that it is a rectangle, then a quadrilateral is a square.
(c) If a quadrilateral is a square, then it is a rectangle.
(d) If it is not true that a quadrilateral is a square, then it is not true that it is a rectangle.

10. The proposition "If a quadrilateral is a square, then it is a rectangle" is true. Its converse "If it is a rectangle, then a quadrilateral is a square" is: 1

(a) It cannot be decided **(b)** False

(c) True **(d)** True only for some values

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the converse of "If a number is divisible by 6, then it is divisible by 3". Is it true? 2

12. Give a counterexample to the statement "The sum of two prime numbers is even". 2

13. How many factors does 64 have? What does the answer tell you about 64? 2

14. Write the converse of "If a quadrilateral is a square, then it is a rectangle". Is it true? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the converse of "If a number is divisible by 6, then it is divisible by 3". Is the converse true? Justify. 3

16. Show that the statement "Every prime number is odd" is false by giving a counterexample. 3

17. Use the converse of the Baudhāyana–Pythagoras theorem to decide whether 15 cm, 36 cm, 39 cm are the sides of a right triangle. 3

OR

Write the converse of "If a quadrilateral is a square, then it is a rectangle". Is the converse true? Justify. 3

18. Show that the statement "If $a < b$, then $a^2 < b^2$" is false by giving a counterexample. 3

OR

Use the converse of the Baudhāyana–Pythagoras theorem to decide whether 16 cm, 30 cm, 34 cm are the sides of a right triangle. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that the sum of any 3 consecutive integers is divisible by 3. Verify for the numbers starting at 8. 5

OR

Prove that the square of an odd integer is odd and write the converse. Is the converse true? Test it with 7. 5

20. Prove that the sum of any 5 consecutive integers is divisible by 5. Verify for the numbers starting at 12. 5

OR

Prove that the square of an odd integer is odd and write the converse. Is the converse true? Test it with 11. 5

END OF QUESTION PAPER — Total: 40 Marks